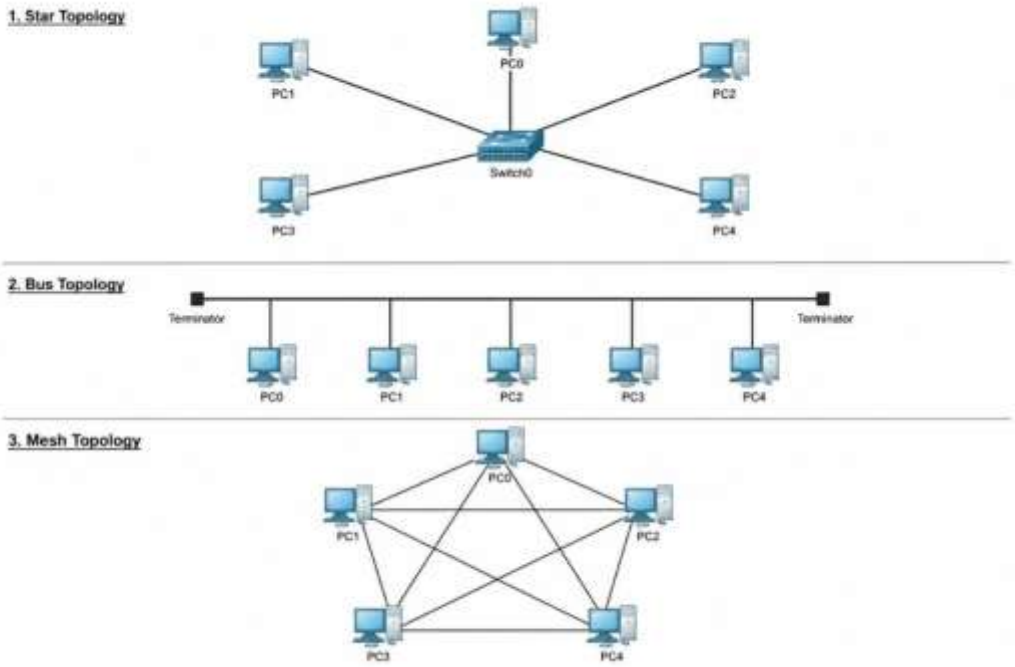
1.

Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

ANS:-



A. Star Topology

- Diagram: Connect all computers to one central switch or router.
- Real-world example: WhatsApp (simplified analogy).
- Explanation: In a star topology, every device connects to a central point. Similarly, WhatsApp users communicate through central servers that handle message delivery.

B. Bus Topology

- Diagram: Connect all computers to one main backbone cable, with terminators at both ends.
- Real-world example: Flipkart (hypothetical internal-network example).
- Explanation: In a bus topology, all devices share one main communication cable. This can illustrate a small local network, although modern Flipkart systems do not necessarily use this topology.

C. Mesh Topology

- Diagram: Connect multiple nodes or servers to one another using several links.
- Real-world example: IPL streaming on JioHotstar (simplified analogy).

- Explanation: Mesh networks provide multiple paths for data transmission. This can help illustrate how distributed systems maintain connectivity if a link fails, although actual streaming infrastructure is more complex.

2.

Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

ANS:-

Comparison of Network Topologies

Topology	**Advantages (Pros)**	**Disadvantages (Cons)**	**Best Use Case**	**App/Service Example**
Star	1. Easy to install. 2. Easy to troubleshoot. 3. Failure of one device does not affect others. 4. Easy to expand.	1. Requires more cables. 2. Central switch failure stops the network. 3. Higher cost than bus. 4. Performance depends on the central device.	Offices, schools, computer labs	Instagram (simplified analogy for client-server communication)
Bus	1. Requires less cable. 2. Low installation cost. 3. Simple for small networks. 4. Does not require a central switch.	1. Backbone cable failure affects the entire network. 2. Difficult to troubleshoot. 3. Performance decreases with heavy traffic. 4. Limited expansion.	Small, temporary or legacy networks	Zomato (hypothetical analogy for a small local network)
Ring	1. Data travels in an organized sequence. 2. Predictable performance under controlled traffic. 3. Can reduce data collisions in token-based rings. 4. Each node gets a fair chance to transmit in a token-based system.	1. A single link or node failure can disrupt a basic ring. 2. Adding devices can be difficult. 3. Troubleshooting can be complex. 4. Delay can increase as data passes through more nodes.	Specialized industrial networks and legacy LANs	Paytm (hypothetical analogy for a ring-based internal network)
Mesh	1. Highly reliable. 2. Multiple paths for data transmission. 3. No single link failure necessarily stops communication. 4. Supports fault tolerance.	1. Expensive to install. 2. Requires many cables and ports in a wired mesh. 3. Complex to configure. 4. Difficult to maintain as the number of nodes grows.	Critical networks, data centers and wireless mesh systems	Instagram or online gaming infrastructure (simplified analogy for interconnected servers)

3.

Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

ANS:-

Suitable Topology for a Multiplayer Mobile Game

Chosen topology: Mesh topology

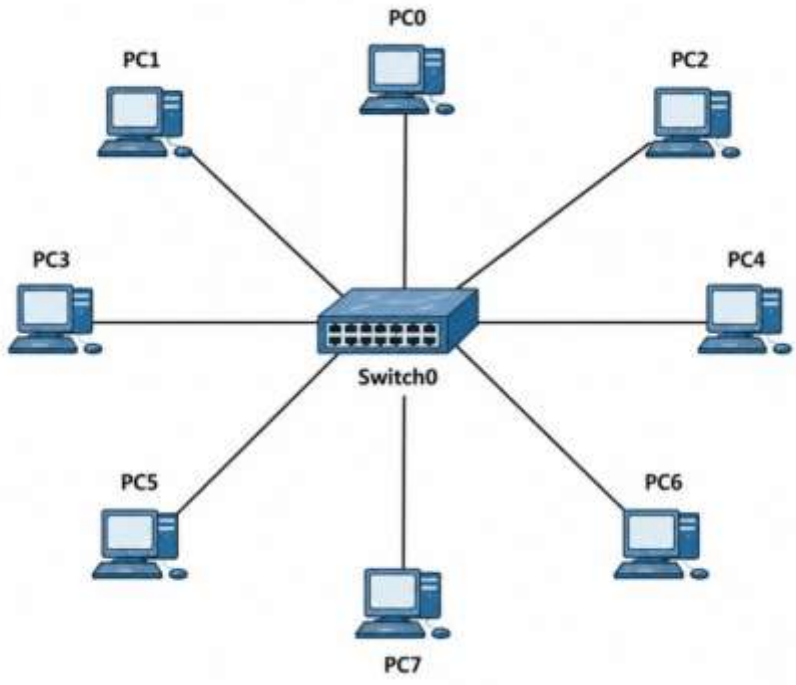
A mesh topology is suitable for connecting game servers because it provides multiple paths for data transmission. It can improve reliability and help maintain connectivity if a link fails. However, a full mesh can be expensive and complex, so a partial mesh is more practical for a large game like PUBG or Free Fire. It balances speed and reliability with a more manageable infrastructure cost.

4.

Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines. Hint: Consider factors like cost, ease of troubleshooting, and future expansion.

ANS:-

Recommended topology: Star topology



Reasoning: Star topology is suitable because it is easy to install, troubleshoot, and expand when the warehouse adds more computers. If one computer or its cable fails, the others continue working. The switch adds some cost, but the improved reliability makes it a practical choice for order processing and inventory updates.